ASSIGNMENT

COURSE NAME:

LIFE CYCLE ASSESSMENT AND SUSTAINABILITY

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This **Life Cycle Assessment (LCA)** is a standardized, **Cradle-to-Grave** methodology that quantifies the potential environmental impacts of iPhone across all stages of its life—from raw material extraction (cradle) to disposal or recycling (grave). The LCA should follow guidelines, for example ISO 14040 series and, for ICT products, supplementary standards from ETSI/ITU.

The two primary components are:

- **Goal and Scope Definition:** Declaration of the purpose of the LCA, the functional unit (e.g., "one iPhone over a 3-year lifetime"), the system boundaries (e.g., raw materials, manufacturing, distribution, use, end-of-life), and any assumptions regarding the data, allocation rules, and impact categories.
- **Inventory Analysis:** This is the phase where all inputs (materials, energy, etc.) and outputs (emissions, waste, etc.) associated with each life cycle stage are compiled and quantified.

This overview gathers details from Apple's official environmental reports (Product Environmental Reports [PER]), peer-reviewed academic studies, and third-party assessments, and focuses on the two key LCA components—Goal and Scope Definition, and Life Cycle Inventory Analysis.

1. Framework for a "Life Cycle Assessment of iPhone"

1.1. INTRODUCTION

1.1.1. Purpose and Rationale:

Explain why assessing the environmental impact of the iPhone is important (e.g., due to high consumer turnover, planned obsolescence, and significant raw material use).

1.1.2. Scope of Study:

Define the functional unit (e.g., one iPhone over a 3-year use period) and establish the system boundaries (from raw material extraction to end-of-life).

1.2. GOAL AND SCOPE DEFINITION

1.2.1. Goals:

i. **Quantification of Environmental Impacts:** The primary aim is to determine the total environmental burden—expressed typically as Global Warming Potential (GWP), among other indicators—across the entire life cycle of the iPhone.

ii. **Hotspot Identification:** Identifying which life cycle stages (e.g., raw material extraction, manufacturing, packaging, use, and end-of-life) contribute most significantly to emissions and resource use.

Functional Unit: One iPhone (128GB configuration) delivering a 3-year service life.

The chosen **functional unit** is fundamental for internal and external comparability. For "one unit of an iPhone (configured as per Apple's current market variant i.e. 15 or 16 series) over a 3- to 4-year operational lifespan" to capture both production and use phase impacts as the iPhone usually is used for a longer period by a consumer.

1.2.2. Scope Details:

The scope of this LCA would be **Cradle-to-Grave** i.e. Raw Material Extraction to End of Life.

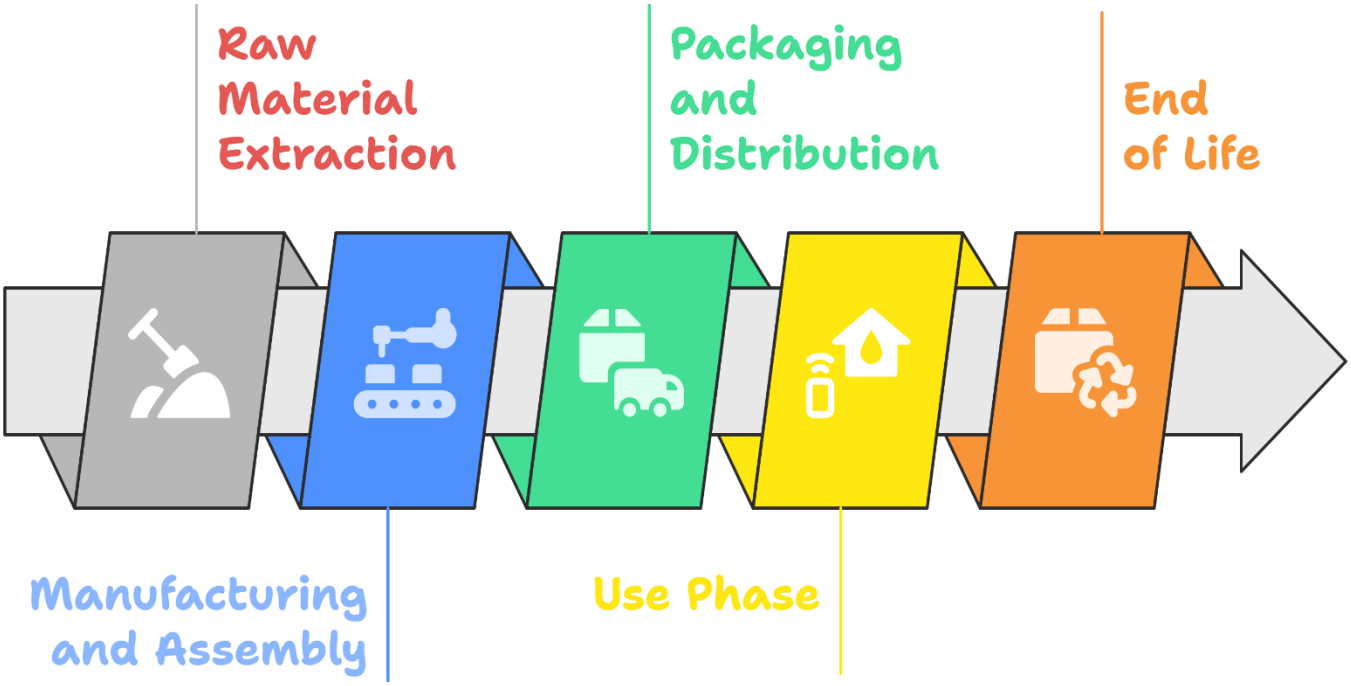
i. **Raw Materials Extraction:** Include the mining and processing of metals such as gold, copper, aluminum, and rare earth elements, as well as production of plastics and glass.

ii. **Manufacturing and Assembly:** Cover the production of key components, such as integrated circuits (ICs), displays, batteries, and the assembly process. Apple's PERs (for example, the iPhone 15 series reports) provide real-time data showing that manufacturing often accounts for approximately 80% of the iPhone's carbon footprint.

iii. **Packaging and Distribution:** Encompass the materials used in packaging (notably Apple's shift to 100% fiber-based packaging with high recycled content) and the energy used in transporting the product.

iv. **Use Phase:** Quantify energy use over the typical lifespan (e.g., periodic battery charging and other direct consumption). Official Apple disclosures often attribute around 15–18% of the total GHG emissions to the use phase; however, independent analyses (such as those from Greenly) suggest that including network and data center emissions could increase the use-phase impact.

v. **End-of-Life (EoL):** Consider recycling, refurbishment, or disposal processes. Although Apple's reports typically indicate that the EoL phase contributes less than 1% of total emissions, it remains an important stage for assessing resource recovery and waste management.



1.3. LIFE CYCLE INVENTORY ANALYSIS

The following elements should be captured:

1.3.1. Raw Materials Extraction and Processing

Materials: A wide array of inputs including metals (e.g., gold, copper, aluminium, rare earth elements) and non-metals (plastics, glass).

Environmental Impacts: High energy consumption and associated greenhouse gas emissions, as well as risks such as soil and water contamination.

For instance, Apple's official reports document that the extraction phase is a significant contributor—up to 61% of the total GHG emissions in some models—as it involves mining, transport, and processing of rare metals [core.ac.uk].

1.3.2. Manufacturing/Assembly

Processes: Transformation of raw materials into high-precision components such as integrated circuits (ICs), displays, batteries, and other parts.

Energy Use: Significant electricity consumption is reported during this phase. According to several Apple Product Environmental Reports (e.g., iPhone 14 Pro, iPhone 16e [apple.com]), production is typically the largest life cycle impact contributor, often accounting for 80% or more of the device's carbon footprint.

Inventory Data: Apple's reports break down emissions by production processes, material and process emissions, and use of supplier low-carbon energy, offering detailed figures (see models from iPhone 15 series [greenly.earth for carbon footprint estimates]).

1.3.3. Packaging and Distribution

Packaging Materials: Apple emphasizes the use of 100% fiber-based packaging and recycled or responsibly sourced wood fibers.

Transport: Distribution involves air, rail, and sea transportation. Although this stage contributes a smaller fraction (around 3–6% for many models), it is still an important aspect of the inventory (data available in Apple's PERs [apple.com]).

1.3.4. Use Phase

Energy Consumption: Battery charging and operational energy use represent a smaller fraction (typically around 15–18% of total emissions in Apple's reports). However, some independent analyses (e.g., from Greenly [greenly.earth]) argue that when network data usage and data center energy are included, the use phase could contribute substantially more.

Functional Considerations: Factors such as average usage time (usually modelled as 3–4 years) and standby versus active consumption are captured.

1.3.5. End-of-Life

Recycling and Disposal: This stage covers the energy and emissions associated with recycling processes, mechanical separation, and any further processing of residual wastes.

Allocation Issues: The method of accounting for end-of-life (EoL) emissions can greatly alter the overall LCA results. Some studies (e.g., [atlantis-press.com] from Ercan et al.) report that EoL typically contributes less than 1% to overall emissions but is still relevant especially regarding resource recovery.

2. Incorporating Real-Time Data from Apple's Disclosures

The most recent ***Product Environmental Reports*** (***PERs***) offer real-time data:

2.1. Carbon Footprint Figures

For example, Apple's disclosures for the iPhone 15 series and iPhone 16 models report unit-level emissions ranging from approximately ***56*** to ***110 kg CO_2-eq***, with production frequently identified as the dominant contributor.

2.2. Recycled Content and Clean Energy

Recent models include over ***25%–30%*** recycled content and report significant reductions in emissions due to the use of supplier low-carbon electricity.

2.3. Use Phase

Apple assumes a ***3-to-4-year lifetime for typical usage***; the reported figures account for controlled charging scenarios. Independent estimates might suggest incorporating additional impact from network use.

This real-time data should back the inventory numbers in each stage of the LCI. For instance, when reporting the energy consumption during manufacturing, we would state it as: "According to Apple's iPhone 15 Product Environmental Report, the production phase contributes approximately 80% of the total carbon footprint, estimated at **56 kg CO_2-eq** for the **128 GB configuration**"

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